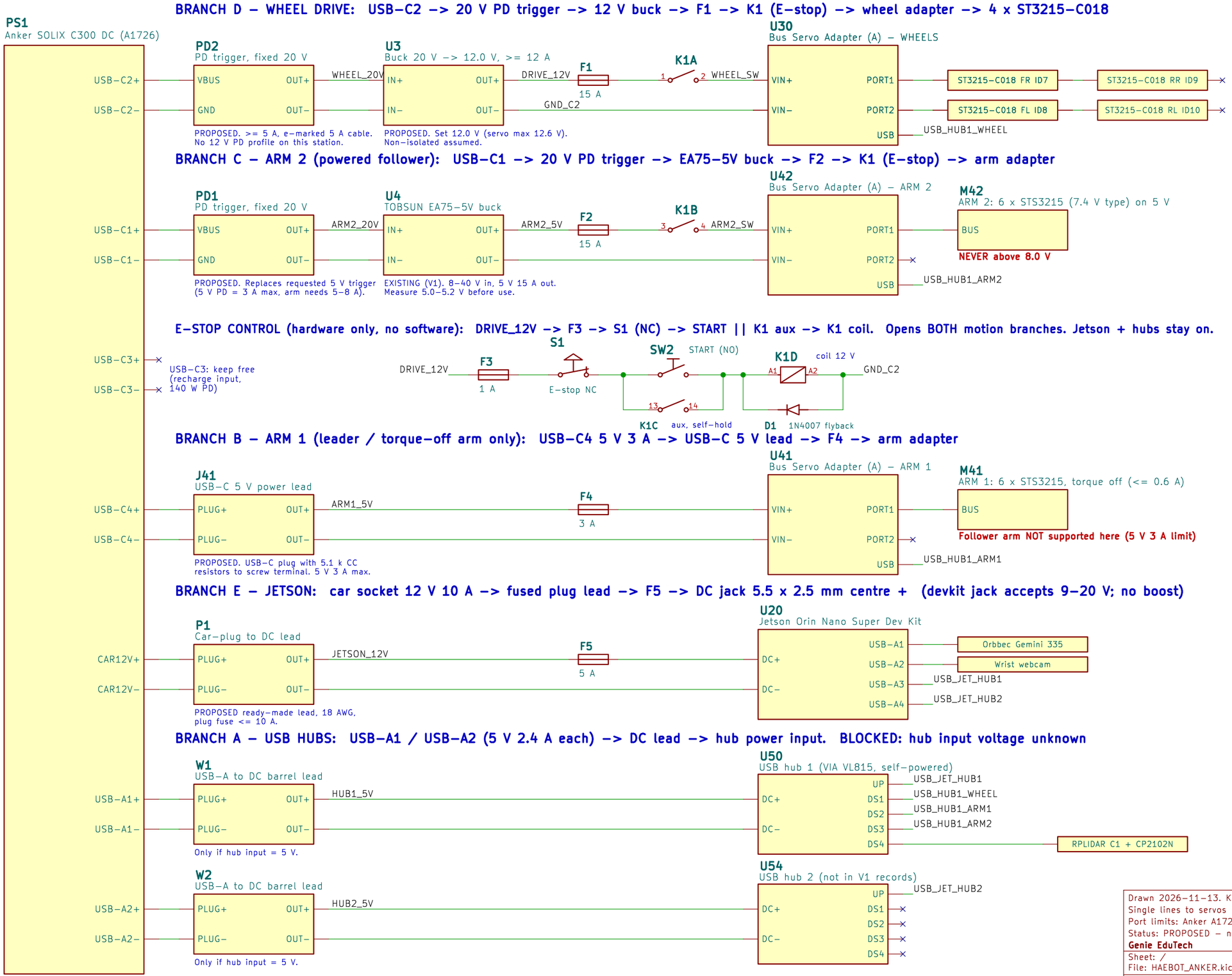
HAEBOT / KIRO V2-A ANKER SOLIX C300 DC POWER SYSTEM

Electrical schematic, Rev A, 2026-11-13. STATUS: PROPOSED - build only after the bench checks in docs/verification_report.md

BRANCH D - WHEEL DRIVE: USB-C2 -> 20 V PD trigger -> 12 V buck -> F1 -> K1 (E-stop) -> wheel adapter -> 4 x ST3215-C018



BRANCH C - ARM 2 (powered follower): USB-C1 -> 20 V PD trigger -> EA75-5V buck -> F2 -> K1 (E-stop) -> arm adapter

E-STOP CONTROL (hardware only, no software): DRIVE_12V -> F3 -> S1 (NC) -> START || K1 aux -> K1 coil. Opens BOTH motion branches. Jetson + hubs stay on.

BRANCH B - ARM 1 (leader / torque-off arm only): USB-C4 5 V 3 A -> USB-C 5 V lead -> F4 -> arm adapter

BRANCH E - JETSON: car socket 12 V 10 A -> fused plug lead -> F5 -> DC jack 5.5 x 2.5 mm centre + (devkit jack accepts 9-20 V; no boost)

BRANCH A - USB HUBS: USB-A1 / USB-A2 (5 V 2.4 A each) -> DC lead -> hub power input. BLOCKED: hub input voltage unknown

ANKER PORT LIMITS (manual A1726)

USB-C1: 5/9/15 V 3 A, 20 V 5 A (100 W)
USB-C2/C3: + 28 V 5 A (140 W)
USB-C4: 5 V 3 A (15 W)
USB-A1/A2: 5 V 2.4 A (12 W each)
USB-A1 + A2 + C4 together: 20 W
Car socket: 12 V 10 A (120 W)
Total output: 300 W
NO 12 V USB-C profile, no PPS listed

PORT MIX USED ON THIS SHEET

C1 + C2 + (A1,A2,C4) + car socket
= manual 4-port case:
C1 30 W | C2 100 W | USB group 20 W
| car 120 W

CALCULATED MARGINS

Wheels: 78 W design < 100 W OK
Wheels: 141 W all-stall > 100 W
Arm 2: 28 W design < 30 W marginal
Arm 2: 44 W burst > 30 W
Jetson: 45 W peak < 120 W OK
Hubs + Arm 1: must stay < 20 W

WARNINGS

1. Never join the + of two ports.
2. Plugging or unplugging any port can renegotiate PD and drop the other outputs for a moment.
3. Set and measure U3 (12.0 V) and U4 (5.0-5.2 V) before any servo is connected.
4. Negatives of all ports become common through USB shields. Motor current must return on its own pair, not through USB.
5. Followers drop when K1 opens.

STATUS WORDS

EXISTING = part on the V1 robot
PROPOSED = not bought, not tested
BLOCKED = needs a hardware fact

Drawn 2026-11-13. KiCad ERC must be run in KiCad 8+ before release
Single lines to servos = 3-wire Molex 5264 bus cable (GND / VCC / SIG)
Port limits: Anker A1726 user manual. Loads: V1 Rev F records (kiro-wiring/docs)
Status: PROPOSED - not bench-verified. See docs/verification_report.md
Genie EduTech

Sheet: /
File: HAEBOT_ANKER.kicad_sch

Title: HAEBOT / KIRO V2-A - Anker SOLIX C300 DC power system

Size: A3	Date: 2026-11-13	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1